

An Intelligent Traffic Light System for Urban Intersections Using Deep Reinforcement

Learning: A Case Study of Lagos Traffic

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CHAPTER ONE: INTRODUCTION

1.1 Background to the Study

Traffic congestion is a major issue in urban environments globally, especially in places that are experiencing rapid urban growth that rely mainly on motorized vehicles for transportation. This problem is a major issue in Lagos, Nigeria with the city topping the likes of Los Angeles, USA and Colombo, Sri Lanka as the worst traffic city mid-2024 (Business Day, 2024). Lagos as a city has a population of over 15 million people with most of them depending on motorized vehicles such as cars, and buses to go about their day, poor public transportation systems and the city not being designed to particularly walkable or sufficient rail transport, particularly make gridlocks more common. These common traffic gridlocks causes substantial economic, environmental, and social consequences.

Traffic congestion's economic impact is staggering. Research shows that traffic congestion costs the global economy billions of dollars annually due to lost productivity, fuel wastage and congestion induced avoidance behaviour where people and businesses avoid an area because they know there would be traffic in that area. Research indicates that traffic congestion cause a huge burden on the city's economy, reports say that residents lose about four hours in traffic congestion daily which causes an estimated 4 trillion naira annual loss in productivity(Nairametrics, 2025)

Beyond economic impact, traffic congestion causes environmental and public health consequences .

Traffic congestion causes people to spend more time in their vehicles, which in turn leads to increased fuel consumption, and consequently results in higher emissions that are harmful to the environment.

These emissions cause significant air pollution in Lagos the transportation sector records PM2.5

These levels are nearly fourteen times higher than what the World Health Organization's

recommended limit, with pollutants like particulate matter, carbon monoxide, sulfur consistently exceeding safe thresholds(Clottey & Ogunlade, 2024)

Traffic light systems play a major part in traffic management globally. Most traffic light systems in Lagos make use of fixed-time controlled strategies. These systems run on schedules and timers, which in itself are usually cheaper and easier to implement. These systems have a fatal flaw: they could be inefficient. This is due to their inflexibility, which makes them inefficient. These could lead to unnecessary delays during low-traffic periods and make congestion even worse during peak hours. They also don't take into account emergency vehicle response and changes in traffic demand.

Artificial Intelligence (AI) and machine learning have come a long way , advancements in technology like these provide an opportunity for us to make improvements in traffic light systems. Reinforcement Learning (RL) allows traffic light systems to learn optimal signal control policies by training in a controlled environment and adapting based on observed conditions. An RL based traffic light system in theory should greatly optimise traffic flow and reduce wait time , which should allow for fewer traffic congestions.

Low-cost embedded hardware such as a Raspberry pi, USB cameras and computer vision techniques make it possible to develop an affordable traffic light prototype .The cameras would be used to detect vehicle density and the Raspberry pi process the data and make decisions based on real time traffic feed.

1.2 Statement of the Problem

The current traffic light system in Lagos usually makes use of fixed time signal switching, which does not take into account real-time traffic variation. This causes increased wait time for vehicles, it could increase congestion. These systems also do not react to emergency vehicles. Emergency vehicles such as an ambulance trying to save a life or a fire truck trying to save lives and property should be given priority, but the current traffic light system doesn't do that. This is why there is a need for a smart traffic light system that can dynamically adjust signal timing based on real-time traffic conditions

using affordable and accessible technology.

1.3 Aim and Objectives of the Project Work

Aim

The aim of this project is to model and evaluate an Intelligent traffic light system that adjusts signal timing based on real time traffic conditions using reinforcement learning.

Objectives

The objectives of this project are as follows:

1. To review current traffic light systems including traditional fixed traffic light systems and existing AI-Traffic light systems.
2. To develop a traffic state representation and reward function that captures input such as vehicle density , wait time , and queue lengths from camera data for reinforcement learning
3. To implement a reinforcement learning algorithm(Deep Q network) to optimize traffic in a simulated environment.
4. To create a realistic traffic simulation in SUMO (Simulation of Urban Mobility) with vehicles, intersections and traffic patterns.
5. To integrate camera based vehicle detection using computer vision techniques.

1.4 Significance of the Project Work

This project contributes to the field of developing efficient transportation systems in a city by using reinforcement learning traffic light systems to control traffic flow in cities. It provides a practical case study on implementing computer vision and artificial intelligence techniques on hardware , which further supports research in traffic management systems.

By reducing traffic congestion this system, if implemented in the real world, could possibly increase productivity , reduce the amount of fuel consumption which in turn reduces the amount of emission. This would in turn bring about a lot of economic and environmental benefits in a city like Lagos which is already plagued with traffic congestion issues.

The project makes use of Raspberry Pi and USB cameras that could serve as a building block in creating more cost effective methods on intelligent traffic light systems that would be able to be deployed in the real world.

1.5 Scope of the Project Work

The project focuses on designing and implementing an intelligent traffic light system on a single intersection in an urban environment. Traffic behaviour is going to be modelled using SUMO and reinforcement learning would be used in optimising the signal timing decisions. A small hardware demonstration which is going to make use of Raspberry pi , USB camera would be used to estimate traffic density and control traffic flow. The project is only going to cover a single intersection, not multiple intersections and it does not involve real world deployment.

1.6 Limitations of the Project Work

The project limitation include:

- 1. Simulation-Based Evaluation:** The intelligent traffic light system training is going to be based in a simulation environment which does not account for real world factors like poor weather conditions which could affect camera ability to detect vehicles, unpredictable drivers etc.
- 2. Single-intersection Focus:** The project focuses on only a single intersection; it does not consider traffic from other intersections, which could be used for better traffic coordination.
- 3. Computational Constraint:** Limited computing resources could reduce training duration of the RL agent, which could cause the policy not to be fully optimised.
- 4. Camera-Based Detection limitations:** Vehicle detection mainly makes use of camera input. Real world factors such as bad weather conditions, low visibility, dusty cameras could reduce vehicle detection accuracy.
- 5. Hardware Prototype limitation:** The hardware being used is limited to a Raspberry pi and USB cameras; they can not actually control a real traffic light network as they are not powerful enough and proper safety and legal approval would be required to test it in a real intersection.
- 6. Temporal scope:** Limited project time would restrict the depth of reinforcement training, which could affect the system performance.

1.7 Definition of Terms

Adaptive Traffic Signal Control:

Traffic signal systems that automatically modify signal timings in response to changing traffic conditions rather than following fixed schedules (Oxford Learner's Dictionary, n.d.).

Agent:

In artificial intelligence, an entity that perceives its environment and takes actions to achieve specific objectives; in reinforcement learning, the agent learns through interaction with the environment (Oxford Learner's Dictionary, n.d.).

Congestion:

A condition in which traffic movement is slowed or obstructed due to an excessive number of vehicles on a road relative to its capacity (Oxford Learner's Dictionary, n.d.).

Deep Q-Network (DQN):

A reinforcement learning algorithm that uses deep neural networks to approximate Q-values, enabling learning of optimal actions in complex and high-dimensional environments (Mnih et al., 2015).

Fixed-Time Traffic Signal:

A traffic control system that operates using predetermined signal timings that do not change in response to real-time traffic demand (Oxford Learner's Dictionary, n.d.).

Green Wave:

A coordinated traffic signal strategy in which consecutive traffic lights are timed to allow continuous vehicle movement along a corridor without stopping (Oxford Learner's Dictionary, n.d.).

Intelligent Transportation Systems (ITS):

Advanced applications that employ information, communication, and control technologies to improve the efficiency, safety, and sustainability of transportation systems (Oxford Learner's Dictionary, n.d.).

Intersection:

A place where two or more roads meet or cross, typically regulated by traffic control mechanisms such as signals or signs (Oxford Learner's Dictionary, n.d.).

Machine Learning:

A branch of artificial intelligence that enables computer systems to learn from data and improve performance without being explicitly programmed for each task (Oxford Learner's Dictionary, n.d.).

Multi-Agent Reinforcement Learning (MARL):

A reinforcement learning framework in which multiple agents interact within a shared environment, learning policies that account for the presence and actions of other agents (Busoniu et al., 2008).

Policy:

In reinforcement learning, a strategy or rule that defines the actions an agent takes in each possible state of the environment (Sutton & Barto, 2018).

Queue Length:

The number of vehicles waiting in line at a traffic control point such as an intersection or signalized junction (Oxford Learner's Dictionary, n.d.).

Reinforcement Learning (RL):

A machine learning approach in which an agent learns optimal behavior through trial-and-error interactions with an environment, guided by reward signals (Oxford Learner's Dictionary, n.d.).

Reward Function:

A mathematical function that assigns numerical values to actions taken by an agent, indicating the desirability of outcomes and guiding learning toward optimal behavior (Sutton & Barto, 2018).

Signal Phase:

A specific part of a traffic signal cycle during which certain traffic movements are permitted to proceed (Oxford Learner's Dictionary, n.d.).

State Space:

The complete set of possible states that an agent can observe in an environment, representing all relevant system conditions (Sutton & Barto, 2018).

SUMO (Simulation of Urban Mobility):

An open-source microscopic traffic simulation software used to model vehicle movement, traffic flows, and signal control in urban road networks (Krajzewicz et al., 2012).

Traffic Throughput:

The rate at which vehicles successfully pass through a road segment or intersection within a given time period (Oxford Learner's Dictionary, n.d.).

Vehicle Detection:

The identification of vehicle presence, count, or characteristics at specific road locations using sensors, cameras, or similar technologies (Oxford Learner's Dictionary, n.d.).

Waiting Time:

The length of time a vehicle remains stationary or delayed in traffic, particularly at intersections or signal queues (Oxford Learner's Dictionary, n.d.).

CHAPTER TWO: LITERATURE REVIEW

2.1 Overview of Traffic Congestion in Urban Areas

This chapter is going to serve as a comprehensive review of existing literature which relates to the field of traffic congestion and intelligent traffic light signal control systems. The aim of this literature review is to examine already existing research, theories, and technological approaches that have been previously proposed in solving traffic congestion in urban environments. This chapter aims to provide

an empirical and theoretical foundation for designing and implementing an intelligent traffic light system through the use of artificial intelligence , by critically analyzing studies relevant to this field.

This chapter specifically focuses on the causes and the impact of traffic congestion, traditional and advanced traffic light systems and the use of artificial intelligence in intelligent traffic light systems, which particularly make use of Reinforcement Learning .The chapter is also going to focus on existing traffic light simulation tools and object detection. Limitations and gaps of previous research are also going to be highlighted . The gaps would serve as necessary motivation and justification for an intelligent traffic light system .

2.1.1 Global Perspective of Urban Traffic Congestion

Road traffic congestion still remains a major problem in urban cities all over the world and this is an issue that is not exclusive to first world, developing, or third-world countries. According to Middle East Economy (2025), the top 3 cities with the highest road traffic in 2025: Lagos (Traffic index: 354.5), Los Angeles (Traffic index: 314.8), Colombo (Traffic index: 292.2). This shows that traffic congestion affects cities across continents and economic levels, truly making this a global issue.

Cities like New York , Paris and Tokyo also experience major traffic congestion issues but factors like: urban planning, cultural attitude and proper public transport system help relieve some of dependency on private vehicles, making congestion more manageable.

2.1.2 Concept and Definition of Traffic Congestion

Traffic congestion in simple terms can be defined as excess travel time or delay beyond free flow of traffic. Traffic congestion also known as “go-slow” or “hold up”in Lagos can be defined as a situation when vehicle overwhelms limited supply of road.It can also be defined as a Congestion can be

defined as the situation when traffic is moving at speeds below the designed capacity of a roadway (Downs, 2004).

2.2 Causes of Traffic Congestion

The Causes of traffic congestion can be categorised into two which are:

2.2.1 Micro-Level Factors of Traffic Congestion

Micro-level factors are factors that relate directly to traffic conditions on the road. These include accidents that occur on the road, a car breaking down, poor driving habits, even phantom traffic jams. These factors disrupt traffic flow and cause delays on road networks.

2.2.2 Macro-Level Factors of Traffic Congestion

Macro levels are structural and socio-economic factors that cause traffic congestion. They are more than just physical things we can easily observe. They include car ownership rates, land use patterns, income levels, infrastructure & network factors. These factors influence travel patterns and how people use the road and long term traffic demand (Rao & Rao, 2012).

2.3 Impact of Traffic Congestion

Traffic congestion has a wide range of impact in urban environments. These impacts range from economic impact to environmental impacts which could affect public health. These impacts highlight the crucial need for efficient traffic light systems in urban environments.

(a) Economic Impact of Traffic Congestion

Traffic congestion causes massive economic costs not just for individuals but also for businesses. Issues like time lost in traffic reduce economic productivity, fuel wastage due to congestion increases fuel expenses for individuals and business, and traffic congestion usually leads to a hike in price of public transportation fees. According to Danne Institute for Research (2021) , “gridlocks make neither naira nor sense”. Their research shows that traffic gridlocks cost both people that use public transport an additional ₦79,039.40 in extra transport fares and it costs people that have their own vehicles around ₦133,978.68 in extra fuel cost yearly. According to the Danne Institute for Research’s “Connectivity and Productivity Report” (July 2023), “traffic congestion costs Lagos about ₦4 trillion annually in lost productivity, wasted fuel, vehicle wear, and missed business opportunities,” a figure the Lagos State Government reaffirmed in June 2025, describing it as a “massive economic drag.” (Business Day, 2025). These economic challenges show why adopting an intelligent traffic control system that optimizes traffic flow and reduces unnecessary delays.

(b) Health & Environmental Impact of Traffic Congestion

Traffic congestion contributes to immense pollution in the urban environment . It produces a lot of noise pollution from car engines running, honking of car horns, sirens and exhaust systems all contribute largely to noise pollution in urban environments. Vehicles caught in traffic congestion also contribute to a lot of emissions of higher levels of greenhouse gases and air pollutants due to vehicles being idle and frequent stop and go motion. The World Health Organisation (WHO) reports that the transport sector is responsible for a large portion of air pollution and greenhouse gas globally. Pollutants such as carbon monoxide, nitrogen oxides, and particulate matter all reduce air quality and contribute to climate change. All these emissions also lead to respiratory issues. An estimated 4.2 million premature deaths are attributed to ambient (outdoor) air pollution.Reducing traffic congestion would therefore reduce the amount of emission vehicles can produce.

(c) Safety & Social impact of Traffic Congestion

Traffic congestion also poses safety and social risks in urban environments. The stop and go motion of traffic congestion increases the risk of rear ended collisions. It also reduces room for evasive maneuvers potentially making some types of accidents more fatal. Another safety concern is that traffic congestion reduces the response speed at which emergency vehicles such as ambulances and fire trucks can get to their destinations to possibly save lives and property. Then social impacts such as road users experiencing more stress and fatigue due to traffic which in turn increases emotional stress, fatigue and health. An intelligent traffic light system would help reduce congestion collisions and it would also reduce the amount of time emergency vehicles need to stay in traffic congestion.

2.4 Conventional Traffic Signal Control Systems

2.4.1 Fixed-Time Traffic Signal Control

Fixed time traffic light system control systems were some of the earliest and widely accepted traffic light signal management systems. In fixed time control, signal phases and the length of green lights are already predefined based on historical traffic data and they remain the same regardless of the real-time traffic conditions. These traffic light system timings are usually determined during the planning stage and calculations are made using traffic flows and mathematical models, when these are done they are then implemented as a static schedule.

One of the most influential works on fixed traffic light signal control was carried out by Webster(1958), he developed a mathematical model to calculate the most efficient traffic light timing . It works by balancing green light time for each direction based on vehicle demand while trying to minimise overall delay. Webster methods simplicity and effectiveness made it relatively easy to implement and the method is still widely made use of today.

The major advantages of fixed traffic signal systems include simplicity to implement,

Minimal hardware requirements and simplicity of operation. Because these systems do not rely on real time sensing or require complex computation, they are usually implemented in regions where resources and technical expertise are limited. Fixed traffic signals are predictable therefore they are beneficial for driver familiarity.

Fixed traffic light systems also have their flaws. One major flaw is their inability to respond to real-time traffic demands which then results in unnecessary long wait times during low traffic periods and may even increase congestion during high traffic periods. Studies have shown that fixed traffic systems fail to account for situations like poor weather ,changes in traffic caused by accidents or just very demanding traffic in general. This is why fixed traffic light systems wouldn't be suitable for highly demanding and unpredictable urban environments . These reasons have led to the development of more advanced traffic light systems.

2.4.2 Actuated and Adaptive Traffic Signal Systems

Actuated and adaptive traffic signals were created to address the shortcoming of fixed time traffic control signals came with, which were that they do not have real-time responsiveness. These systems usually make use of sensors such as inductive loop detectors, infrared sensors and cameras to collect traffic data and then with that data control traffic signals. Unlike fixed-time systems, actuated and adaptive controllers control traffic systems based on input received such as vehicle presence ,traffic density or queue lengths.

Actuated traffic signals work simply by extending or terminating green light at intersections based on sensor input. This approach greatly improves efficiency compared to the fixed time traffic control signals, this system usually is most effective in moderate traffic areas. They are usually less expensive than adaptive traffic signal systems.

Adaptive traffic control refers to a sophisticated system designed to optimise traffic flow by dynamically adjusting signal timings based on real-time data (OPTraffic, 2025). Adaptive systems use real-time data collected from traffic using sensors to evaluate traffic. These systems analyse factors like vehicle count, queue length, and traffic density. Based on these inputs they adjust traffic signals.

The Split Cycle Offset Optimisation Technique (SCOOT) and the Sydney Coordinated Adaptive Traffic System (SCATS) systems are both adaptive traffic control signal systems. They use real time traffic measurements to adjust incrementally signal timing , offsets and cycle lengths in an urban environment. The system continuously accesses traffic conditions and continuously makes adjustments to reduce stops and delays.

Actuated and adaptive traffic systems have the upper hand over fixed traffic light systems because they reduce delays, improve traffic flow and are more effective at handling dynamic traffic demands. They usually involve a significant amount of sensor infrastructure, costly installation and maintenance costs and centralized control systems, which can be a hassle to deploy in resource-constrained environments. They are also not really adaptable; this is because their adaptability is usually rule-based and dependent on pre-defined control logic, which may hinder their capacity to manage complex and highly dynamic traffic scenarios

These limitations have resulted in growing research interest in artificial intelligence based traffic signal control methods, specifically those based on reinforcement learning techniques, which try to learn optimal signal control policies from traffic interactions, instead of predefining rules or models.

2.5 Intelligent Transportation Systems (ITS)

2.5.1 Overview of Intelligent Transportation Systems

Intelligent Transportation Systems (ITS) in simple terms can be defined as an interconnected network of technologies designed to facilitate movement of people and goods. ITS makes use of sensing technologies, data processing , communication networks, and control systems to enable efficient decision-making in transport environments. ITS helps determine smarter route guidance , reduce fuel consumption, reduce traffic congestions, supports energy independence and sustainability. Contribute to healthier, more livable cities. Not just on land but also on sea and air.(International Organization for Standardization [ISO], n.d.).

Information and Communication Technologies (ICT) play a crucial role in helping traffic management by helping in enabling real time data collection and analysis. Technologies like traffic sensors, surveillance cameras, GPS-enabled devices and wireless networks allow for road monitoring and proper traffic management. They help in supporting adaptive traffic light systems, congestion predictions, incident prevention and accident response. ITS has massive potential in improving traffic efficiency and road capacity without needing to increase costly physical road infrastructure .

Globally ITS have been implemented in developing and developed countries. Countries like the United State of America, Singapore , China , Netherland and UAE are some of the countries that lead in implementing ITS solutions for traffic signal coordination, electronic toll collection, and smart public transportation. Lagos Nigeria is also implementing Intelligent Transportation Systems, and smart cameras are being used in locations like Ikeja and Third Mainland Bridge to detect traffic violations. Lagos still has a long way to go in implementing Intelligent Transportation Systems in reducing traffic congestions using ITS.

2.5.2 Role of Artificial Intelligence in Traffic Management

Any system capable of simulating human intelligence and thought processes is said to have “Artificial Intelligence” (AI) (IBM, 2022).

Artificial intelligence has revolutionized traffic management, it has the ability to predict traffic density, detect potential congestion and even predict how events like road closure or accidents could affect traffic flow. Beyond predictions AI can also help with optimising traffic control in real time. AI based traffic prediction models significantly outperform traditional fixed traffic signal systems, they have the ability to predict traffic congestion up to 30 minutes in advance. This allows for external interventions to be carried out such as adjusting traffic signals and re-routing vehicles, which could reduce overall traffic time by up to 20% (REST Publisher, 2024).

AI already plays a huge role in people's day to day movement. When you use google maps and google gives you an estimated arrival time, google aggregates location data to understand traffic on the road. This is fine during a current point in time, it does not account for what traffic would look like in 10, 20 ,50 minutes. To predict how traffic would be in future google uses historic traffic patterns. These historic patterns are combined with current traffic data using machine learning models to generate more traffic ETA predictions. This shows that AI is already playing a huge role in traffic management.

The AI-powered traffic management systems currently include prediction, optimization, and real-time control to develop intelligent and adaptive traffic solutions. These systems are dynamically implemented to assign green signal times at intersections and minimize congestion, delays, and vehicle emissions using genetic algorithms, particle swarm optimization, and deep reinforcement learning. Contrary to the conventional permanently fixed or rule-based controllers, AI systems can react to real-time sensor information in the span of milliseconds, and traffic lights can be adjusted immediately to varying road conditions. When implemented with edge devices like Raspberry Pi, they work without the input of the cloud making them reliable even when the network goes down. It has

been demonstrated that prediction, optimization and real-time execution in a closed-loop system may enhance traffic throughput up to 25 to 35% compared to standard traffic control systems (REST Publisher, 2024).

2.6 Reinforcement Learning in Traffic Signal Control

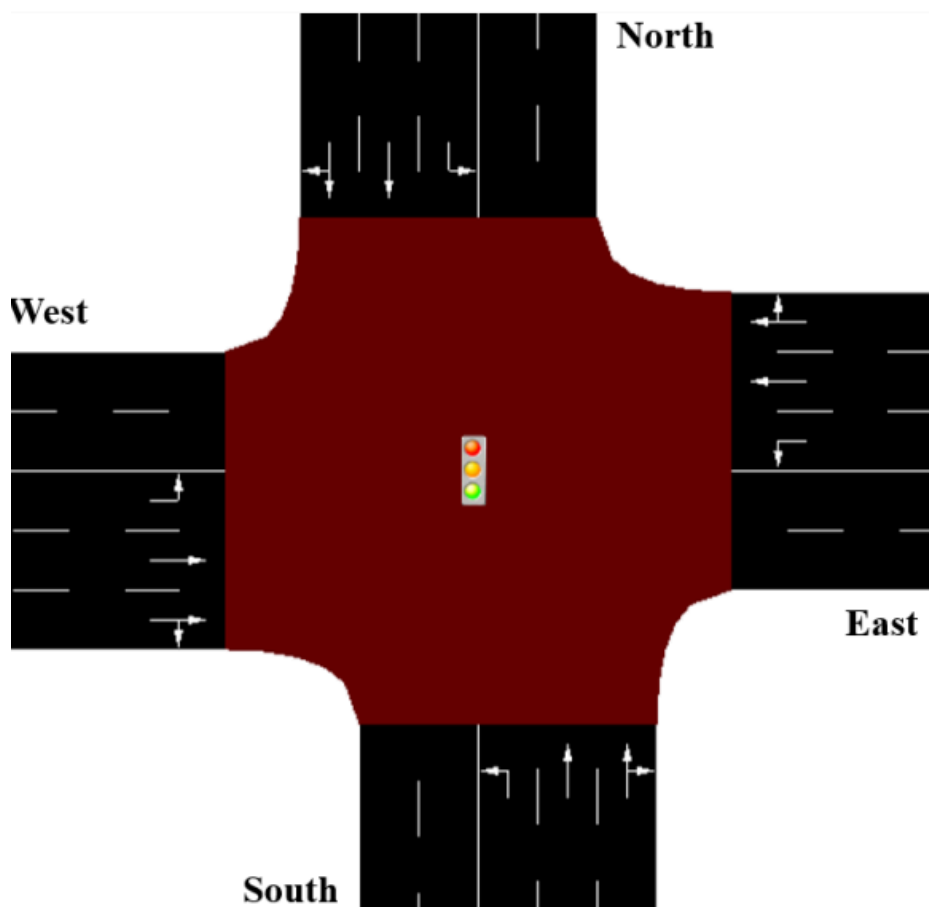


Fig 1 Layout of the signalized intersection under study

2.6.1 Fundamentals of Reinforcement Learning

Reinforcement learning (RL) is a type of machine learning process in which autonomous agents learn to make decisions by interacting with their environment (IBM, nd)

Reinforcement Learning could regulate traffic by treating traffic signals as a sequential decision making problem within a Markov Decision Process (MDP) framework. The agent in this case would be the signal controller observing the intersection and deciding phase action. The environment for training would encompass a traffic network modeled through simulation (e.g SUMO), responding to agent actions with evolving traffic trajectories.

Reinforcement learning while formulating an Intelligent traffic light system, the problem would be defined by $\langle \mathbf{S}, \mathbf{A}, \mathbf{R} \rangle$. The S_t represents **STATE** which represents real-time traffic conditions and is defined by a 12 dimensional vector $S_t = [q_1, q_2, q_3 \dots, q_{12}]$ where q_i represents the number of halting vehicles in each lane (meaning vehicle speed ≤ 0.1 m/s) so the AI system would see something like $S = [8, 7, 6, 0, 0, 1, 9, 10, 8, 0, 0, 0]$. The $\mathbf{A}$ represent **Action space** is denoted as $\mathbf{A} = \{\Phi_1, \Phi_2, \dots, \Phi_8\}$

This corresponds to the eight traffic signal phase at the intersection as seen in fig 1 above, where selecting the same phase consecutively increases duration by a fixed amount Δ , while selecting a different phase initiates a yellow light transition into the new phase. The $\mathbf{R}$ represents **Reward** this function is designed to encourage reduction in congestion and it is the change in total cue length

between consecutive time steps, given by $R_t = L_t - L_{t+1}$ where, $L_t = \sum_{i=1}^{12} q_i$; thereby a reduction in stopped vehicles grants a positive reward.

From these definitions **Policy** $\pi(a/s)$ is learned using a Q-network that approximates the optimal action-value function $Q^*(s, a)$, guiding the agent to select actions that maximize the expected cumulative discounted reward $E[\sum_{k=0}^{\infty} \gamma^k R_{t+k}]$, thereby minimizing queue lengths and delays at the intersection (Tislenko & Kisilev, 2024).

2.6.2 Deep Reinforcement Learning Approaches

Deep Reinforcement Learning builds upon the tabular techniques with the function approximation of a neural network that operates on traffic states of any dimensionality. Deep Q-Networks (DQN) are the basic architecture which parameterizes the action-value function.

Deep convolutional and fully connected layers $Q(s, a; \theta)$, (Mnih et al., 2015). Input states are presented as picture-like representations of intersection queues, or as a vectorized occupancy of lanes, however with the output being Q-values of each of all possible phase actions.

DQN uses experience replay and target network stabilization to reduce dependency on sequential traffic data. The loss function minimizes temporal difference error:

$L(\theta) = E[(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta))^2]$ where θ^- denotes frozen target parameters updated periodically.

Deep architecture thrives in complex environments, because it automatically extracts structured features obtained from raw sensor signals. Unlike hand-crafted heuristics struggling with non-stationary traffic distributions, DQN discovers phase selection strategies adapting to diurnal patterns, incidents, and demand fluctuations. Case studies report 28% average delay reduction versus fixed-time baselines and 15% improvement over actuated control in SUMO simulations. The approach scales to realistic intersection geometries by leveraging transfer learning from simulation to deployment.

2.6.3 Single-Agent vs Multi-Agent RL Traffic Control

Single-agent RL treats intersections as an MDP (Markov Decision Process), which simplifies state action space when enabling converging rapid policy. The controller observes queue states and then

selects phases , which is suitable for a single intersection or loosely coupled network. Its computational complexity scales linearly with lanes/phases, it facilitates real-time inference on embedded systems like Raspberry pi 4. Single-agent formulation usually ignores downstream spillback effects, potentially creating less optimal network flows.

Multi-agent RL models represent arterial systems as distributed systems in which intersection agents implicitly organize themselves based on traffic dynamics. Models such as MADDPG have centralized critics that assess combined actions and decentralized agents that implement local policies (Lowe et al., 2017). Sanctions: a reward shaping approach with neighbor queue information leads to coordination with a green wave.

Complexity trade-offs usually benefit single-agent approaches especially for prototype development and resource-constrained deployment. Multi-agent systems exhibit non-stationarity as co-learners modify the environment, requiring sophisticated credit assignment and extended training horizons. Empirical comparisons demonstrate single-agent DQN converging 3-5x faster than multi-agent counterparts while achieving 85-90% of coordinated performance in isolated settings.

The single-agent appropriateness is ideal to the scope of this project: the deployment of Raspberry Pi limits the availability of computing power, SUMO validation is limited to a single intersection, and the modular design allows extending the system to multiple agents. The methodology is cognitive and demonstrable in practice, dealing with Lagos intersections that are not covered within a network of communication infrastructure.

2.7 Traffic Simulation Tools for Traffic Signal Research

2.7.1 Overview of Traffic Simulators

Microscopic traffic simulators are essential platforms that allow for simulating and training algorithms under supervised environments . PTV VISSIM uses proprietary car-following

models optimized for western traffic patterns, which offers sophisticated driver behavior calibration but is not open source so it requires substantial licensing costs (PTV Group, 2023). Aimsun Next integrates mesoscopic and microscopic simulation with real-time data assimilation, it excels in network-wide analysis though its complexity deters academic adoption (TSS-Transport Simulation Systems, 2022).

SUMO (Simulation of Urban Mobility) is a predominant open-source alternative, which supports diverse traffic regimes including mixed vehicle types which are usually prevalent in developing cities. Its permissive licensing and Python integration allows it to be a good option for reinforcement learning experimentation, which makes SUMO a de facto standard for signal control research.

2.7.2 SUMO (Simulation of Urban Mobility)

SUMO gained popularity because it has a comprehensive feature set and a very active academic community. Released in 2001, the simulator was able to model vehicle trajectories using Intelligent Driver Model for car-following and MOBIL for lane-changing, accurately replicating emergent congestion phenomena (Krajzewicz et al., 2012). Networks import from OpenStreetMap or NETCONVERT, generating realistic intersection geometries with right-of-way rules and phase definitions.

The open-source MIT license has removed the monetary obstacles and TraCI (Traffic Control Interface) allows the TraCI to communicate with the external controllers in a real-time bidirectional manner. Python bindings can observe states (lengths of queues, speeds) and take action (phase changes) with 10Hz simulation rates. Calibration processes are comparable with empirical flow profiles by adjusting demand and optimization of route distributions.

Lopez et al. (2018) showed the flexibility of SUMO in intermodal operations, where it is connected to the models of public transport and pedestrians, which are critical in analyzing urban environments. The simulator is scalable with single intersections for prototype validation through city-scale networks for policy evaluation.

2.8 Vehicle Detection and Sensing Technologies

2.8.1 Traditional Sensors

Inductive loop detectors still dominate infrastructure-based sensing; they work by embedding wire coils in pavement to measure inductance perturbations from metallic vehicle chassis (Federal Highway Administration, 2005). Dual-loop configurations estimate speeds and detection with about 95% accuracy under good conditions, pavement cuts and maintenance sometimes disrupt functionality. Infrared beam-break sensors are a non-intrusive alternative, they detect axles passing across roadways but usually struggle with multi-vehicle detection.

Microwave radar sensors project Doppler-shifted radio waves, classifying vehicle presence, speed, and length regardless of lane position. These active systems penetrate adverse weather but require precise calibration to mitigate false detections from adjacent lanes (Ki & Baik, 2006).

2.8.2 Camera-Based Vehicle Detection

Computer vision makes use of physical sensors through non-intrusive overhead monitoring.

Viola-Jones (2001) introduced Haar cascade classifiers which enabled rapid pedestrian detection, the foundation of which was used for subsequent vehicle tracking algorithms. Modern single-shot

detectors like YOLO process entire frames in one forward pass, regressing bounding boxes and class probabilities simultaneously (Redmon et al., 2016).

YOLOv8-nano variants achieve around 37.3% mAP on COCO at 80 FPS on consumer GPUs, with TensorFlow Lite quantization allowing Raspberry Pi deployment at 8-12 FPS. Virtual detection zones and Kalman filter tracking count vehicles crossing predefined lines, obviating physical installation. Cost-effectiveness stems from leveraging existing CCTV infrastructure, though performance degrades under glare, heavy rain, or nighttime conditions without IR illumination.

2.9 Review of Related Works

El-Tantawy et al. (2013): Multiagent Reinforcement Learning Framework

Authors: El-Tantawy, S., Abdulhai, B., & Abdelgawad, H.

Methodology: Parameterized MARL with minimax-Q learning across coordination graphs.

Tools: Custom microscopic simulator with dynamic stochastic assignment.

Key Findings: 35% delay reduction vs. actuated control; scalable to 20+ intersections.

Limitations: Assumes perfect state observation; computation-intensive training.

Van der Pol & Oliehoek (2016): Coordinated Deep RL

Authors: Van der Pol, E., & Oliehoek, F. S.

Methodology: Value-Decomposition Networks coordinating DQN agents.

Tools: SUMO with 4x4 grid network.

Key Findings: 22% improvement over independent DQN; emergent coordination.

Limitations: Centralized training infeasible for large networks; sensitive to reward scaling.

Wei et al. (2018): IntelliLight

Authors: Wei, H., Zheng, G., Yao, H., & Li, Z.

Methodology: Light-weight DQN with hierarchical phase selection.

Tools: SUMO + real Hangzhou data.

Key Findings: 18% delay reduction; 5x faster than PressLight.

Limitations: Relies on historical data calibration; limited to arterial networks.

2.10 Research Gaps Identified

Available literature indicates that implementation of intelligent traffic lights in urban developing cities is already in place . The cost of hardware is high which does not allow for wide adoption; training and server-grade inference powered by GPUs are also out of reach . Most other studies give a disproportionate focus on multi-intersection coordination and they usually overlook possible single-junction solutions that can be instituted in a phased manner.

The inapplicability of current models to emerging urban conditions appears in the form of Western-biased assumptions, i.e., homogenous traffic behaviour, well-developed communication systems, and sensor networks that are already installed. Multi-agent reinforcement learning (MARL) systems have an intricate design, which assumes their connection by the network which is not common in most emerging markets.

This project provides an answer to these gaps by deploying Raspberry Pi 4, performing the detection using YOLO and utilizing the current cameras, as well as using single agent DQN trained on edge inference. The strategy focuses on price, compatibility with infrastructure and proven performance on limited resources.

2.11 Summary of Literature Review

The literature demonstrates the progressiveness of traffic signal systems, starting with rigid fixed-time systems through to actuated and adaptive progressive, and now reinforcing learning based traffic systems .Reinforcement learning, particularly DQN variants, consistently outperforms traditional methods by 15-35% across diverse metrics. SUMO provides robust simulation validation, while computer vision enables cost-effective sensing.

However, deployment barriers persist in resource-constrained settings. Based on the reviewed literature, the need for a low-cost, reinforcement learning-based intelligent traffic light system suitable for developing urban environments is evident. Chapter Three presents the proposed methodology addressing these identified gaps.

CHAPTER THREE: SYSTEMS ANALYSIS, DESIGN AND PRODUCTION

3.1 Introduction

This chapter provides the systems analysis, design and production process of the proposed intelligent traffic light system. It is going to explain how the system is analyzed, how the design decisions were made, and how the overall solution is going to be implemented to address traffic congestion issues highlighted in the earlier chapters. The chapter covers the analysis of the existing traffic control system, system requirements, architectural design, reinforcement learning model design, traffic simulation setup, and hardware implementation approach.

3.2 Analysis of the Existing Traffic Light System

3.2.1 Description of the Existing System

Traffic light systems in Lagos use fixed-time control strategies. These systems signal timing that have already been defined and they do not adapt to real-time traffic congestion. They run on the same schedule regardless of vehicle density, congestion levels or even unexpected traffic events .

The existing system usually does not make use of smart sensing or automated decision-making. The timing plan of traffic lights is also not updated frequently, and in most cases, manually, and that is why this is ineffective during busy hour traffic jams, accidents or emergency traffic.

3.2.2 Limitations of the Existing System

The major drawbacks of the systems are:

- The system is unable to respond to real-time traffic issues
- Increased wait time for vehicles during low-demand periods
- Weak performance during high congestion periods
- Emergency vehicles do not get priority

These shortcomings aid congestion, increases fuel consumption, and environmental pollution, particularly in a high-dense city like Lagos.

3.3 Analysis of the Proposed Intelligent Traffic Light System

3.3.1 Overview of the Proposed System

The system consists of an intelligent traffic light control system that operates using reinforcement learning. It adjusts the traffic signal phase dynamically based on input data from a camera. SUMO would be used to train and test the system before the system is deployed on embedded systems.

The system is going to take signal control as a sequential decision making problem and the traffic light controller (agent) is going to switch the signals with the goal of making wait time and congestion as few as possible .

3.3.2 Functional Requirements

The functional requirements of the proposed system are:

- Ability of cameras to detect vehicle and also amount of vehicles in a queue
- Ability to model traffic conditions as a state during reinforcement learning
- Ability to choose the best signal phase
- Ability to update signal timing based on traffic demand
- Ability to simulate traffic in a controlled environment

3.3.3 Non-Functional Requirements

The non-functional requirements are:

- Reduced computational cost so it could be deployed on Raspberry pi
- Scalability to allow future expansion across multiple intersections
- Cost-effectiveness using affordable sensors and hardware

3.4 System Architecture and Design

3.4.1 High-Level System Architecture

The proposed system consists of four major components:

1. Traffic simulation environment (SUMO)
2. Reinforcement learning agent (DQN)
3. Vehicle detection (computer vision)
4. Embedded hardware (Raspberry Pi)

The communication between these components forms a closed-loop control system that allows traffic data to be sensed, processed, learned from, and acted upon in real time.

3.4.2 System Workflow

Here is the system workflow:

1. Vehicles approach the intersection and they are detected by the camera
2. Information such as traffic length is extracted
3. The reinforcement learning agent observes the current state
4. The optimal signal phase is chosen

5. Traffic signals are then updated

6. The environment is a source of feedback on a reward basis

This cycle repeats itself continuously, this then allows the system to make better decisions over time.

3.5 Reinforcement Learning Model Design

3.5.1 State Representation

The state of traffic environment is modelled as a vector comprising of the queue length of all incoming lanes in the intersection:

$$S_t = [q_1, q_2, q_3, \dots, q_{12}]$$

q_i represents the number of stopped vehicles (speed ≤ 0.1 m/s) on lane i .

3.5.2 Action Space

The action space comprises of fixed traffic signal phases:

$$A = \{\phi_1, \phi_2, \phi_3, \dots, \phi_8\}$$

Every action is equated to the activation of a certain phase of signal. The consecutive use of the same phase increases the green time, whereas the alternate use of the phases adds a transition of yellow.

3.5.3 Reward Function

The rewarding operation is meant to reduce congestion by promoting reduction of the queue length:

$$R_t = L_t - L_{t+1}$$

and L_t total queue length at time t . A decrease in queue length is a positive reward, which brings the agent to congestion-reducing policies.

3.5.4 Learning Algorithm

Deep Q-Network (DQN) is going to be used to approximate the optimal action-value function $Q^*(s, a)$. This model makes use of experience replay and a target network to stabilize training and improve convergence in non-stationary traffic environments.

3.6 Traffic Simulation Environment

3.6.1 SUMO Simulation Setup

SUMO (Realistic intersection geometry

- Multiple incomingSimulation of Urban Mobility) is used to simulate traffic behaviour at an intersection. The simulation includes:
 - lanes
 - Fluctuating traffic demand
 - Signal phase definitions

Traffic flows on SUMO are able to simulate peak on and peak off traffic, similar to traffic patterns in Lagos.

3.6.2 Combination with Reinforcement Learning.

The TraCI interface is used to conduct real-time communication between the reinforcement learning agent and the simulator of SUMO. Statuses of traffic are obtained at SUMO and signal control actions are performed dynamically in the course of simulation.

3.7 Vehicle Detection and Sensing Design

3.7.1 Camera-Based Detection Approach

Vehicles are detected using computer vision techniques. A camera observes incoming lanes and YOLO is used to identify and count the vehicles in each lane .

3.7.2 Advantages of Camera-Based Sensing

- Simplicity to install
- Cost-effective compared to loop detectors
- Flexible deployment

- Compatible with existing CCTV infrastructure

3.8 Hardware Design and Implementation

3.8.1 Hardware Components

The hardware prototypes are:

- Raspberry Pi 4
- USB camera
- Traffic signal LEDs (for demonstration)
- Power supply
- Model cars

3.8.2 Role of the Raspberry Pi

The Raspberry Pi is used for edge computing and its roles are:

- Processing the cameras input

- Running inference using the trained DQN model
- Controlling signal phase changes

Edge deployment allows for low latency and continuous operation without needing to depend on the cloud.

3.9 System Production and Implementation Process

The system development is as follows:

1. SUMO Traffic environment modeling.
2. Design and training of reinforcement learning agents.
3. Integration of vehicle detection models.
4. In simulation, system testing.
5. Hardware prototype deployment

All the stages were tested separately and then the system was integrated completely.

3.10 Ethical, Safety, and Practical Considerations

The system will only be carried out in simulation and will be used to demonstrate prototypes. There is no physical traffic signal infrastructure which is controlled. The implementation would be required to comply with safety, legal approval and regulatory compliance before it is implemented in the real world.

3.11 Summary of Chapter Three

This chapter introduced the systems analysis, design and the production process of the intelligent traffic light system proposed. The chapter described the drawbacks of the current systems, the design of reinforcement learning model, simulating environment, sensing and hardware implementation. The second chapter provides the result of system implementation, testing, and performance evaluation.

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